

WELL COMPLETION REPORT

Synthetic Field Cluster — SYN-Assam-Duliajan (Upper-Assam-type setting)

1. WELL IDENTIFICATION

Well ID	W007
Well Name	SYN-Assam-Duliajan-007
Field / Cluster	SYN-Assam-Duliajan
Area	Synthetic Assam Field Cluster (Duliajan-type operating area)
Latitude	27.431355 N
Longitude	95.294900 E
Well Status	Completed
Well Role	Historical (offset reference well)
Spud Date	19-Jun-2021
Drilling Completed	30-Jun-2021
Report / Completion Date	04-Jul-2021
Total Depth (Driller)	3,400 m
Primary Formation	Sandstone
Formations Penetrated	Claystone / Shale / Sandstone / Limestone
Rig	SYN-RIG-E2000-01
Drilling Days	12 days
Recorded Events	3
Recorded NPT	74.0 hours

2. MACHINE-READABLE HEADER BLOCK

Document Type: Well Completion Report
Well ID: W007
Well Name: SYN-Assam-Duliajan-007
Field: SYN-Assam-Duliajan
Latitude: 27.431355
Longitude: 95.294900
Total Depth: 3,400 m
Primary Formation: Sandstone
Well Status: Completed
Spud Date: 2021-06-19
Event Count: 3
Total NPT: 74.0 hours

PROTOTYPE TRAINING RECORD. Synthetic demonstration document prepared for the eRTMAC-NWIS prototype (SIH PS 26121). The well, coordinates, formations, events and measurements described here are generated for software demonstration and are not an operational record of any company. Location names are used for interface readability only.

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3. WELL SUMMARY

3.1 Objective

Well W007 (SYN-Assam-Duliajan-007) was drilled to provide an additional development penetration of the primary sandstone objective and confirm the area formation sequence. The well was drilled vertically from surface to a driller's total depth of 3,400 m within the SYN-Assam-Duliajan synthetic cluster.

3.2 Drilling Interval

Surface to 3,400 m, completed in 12 rotary drilling days between 19-Jun-2021 and 30-Jun-2021. Average overall penetration rate for the well was 11.8 m/hr on a 24-hour basis.

3.3 Formations Encountered

The well penetrated 4 formation units: Claystone (0–764 m), Shale (764–1,841 m), Sandstone (1,841–3,025 m), Limestone (3,025–3,400 m). The primary objective formation was Sandstone.

3.4 Major Operational Events

A total of 3 reportable events were recorded, comprising 2 × Stuck Pipe, 1 × Casing Issue. Aggregate non-productive time was 74.0 hours. Event detail is given in Section 6.

3.5 Casing Operations

3 casing strings were run: 13-3/8" at 765 m, 9-5/8" at 1,840 m, 7" at 3,400 m. All strings were cemented as detailed in Section 7.

3.6 Completion Status

Well status at report date: **Completed**. The well was completed on the primary sandstone objective and handed over for production testing.

4. OFFSET-WELL CONTEXT

The following synthetic wells of the same cluster lie within short offset distance and share a comparable formation sequence. They are the primary comparison set for offset-well analysis.

Well ID	Well Name	Distance (km)	TD (m)	Primary Formation	Events
W002	SYN-Assam-Duliajan-002	4.73	3,450	Sandstone	7
W004	SYN-Assam-Duliajan-004	5.79	3,590	Sandstone	2
W010	SYN-Assam-Duliajan-010	6.93	3,260	Sandstone	7
W001	SYN-Assam-Duliajan-001	7.62	3,470	Sandstone	4
W005	SYN-Assam-Duliajan-005	9.05	3,560	Sandstone	5
W009	SYN-Assam-Duliajan-009	10.80	3,180	Sandstone	6

5. FORMATION / DEPTH RECORD

Formation tops and bases as recorded from cuttings and log correlation. Depths are driller's depths below rotary table.

Unit	Formation	Top Depth (m)	Bottom Depth (m)	Thickness (m)	Remarks
1	Claystone	0	764	764	Routine drilling, no reportable event
2	Shale	764	1,841	1,077	Routine drilling, no reportable event
3	Sandstone	1,841	3,025	1,184	Stuck Pipe, Casing Issue recorded (3 events)
4	Limestone	3,025	3,400	375	Routine drilling, no reportable event

Formation: Claystone Top: 0 m Bottom: 764 m
Formation: Shale Top: 764 m Bottom: 1841 m
Formation: Sandstone Top: 1841 m Bottom: 3025 m
Formation: Limestone Top: 3025 m Bottom: 3400 m

6. DRILLING PARAMETERS BY FORMATION

Representative values summarised from the depth-indexed parameter record for this well. Mud weight is reported in ppg, rate of penetration in m/hr, rotary speed in rpm and weight on bit in klbf.

Formation	Interval (m)	Mud Wt (ppg)	MW range	ROP (m/hr)	RPM	WOB (klbf)
Claystone	0–764	9.55	9.1–10.1	32.5	142	7.8
Shale	764–1,841	10.95	10.2–11.7	16.3	127	16.9
Sandstone	1,841–3,025	12.36	11.5–13.3	15.9	113	24.7
Limestone	3,025–3,400	13.79	13.3–14.1	8.2	94	34.0

6.1 Mud Programme Summary

Water-based inhibited mud was used throughout. Mud weight was raised progressively with depth from approximately 9.6 ppg in the upper section to 13.8 ppg at total depth. No loss-related mud weight reductions were required.

6.2 Bit / Hydraulics Note

Rotary speed was held in the 121 rpm range on average with weight on bit averaging 19.6 klbf. Penetration rate declined with depth from 32.5 m/hr in the upper hole to 8.2 m/hr at total depth, consistent with the area trend.

7. DRILLING EVENTS RECORD

All reportable events for this well, in depth order. Each event is recorded against the formation interval containing the event depth.

Event ID	Depth (m)	Formation	Event Type	Severity	Duration (hr)
E0040	2,317	Sandstone	Stuck Pipe	High	28.2
E0041	2,823	Sandstone	Stuck Pipe	Medium	11.4
E0042	2,852	Sandstone	Casing Issue	High	34.4

Well ID: W007 | Event Type: Stuck Pipe | Depth: 2317 m | Formation: Sandstone | Severity: High | Duration: 28.2 hours

Well ID: W007 | Event Type: Stuck Pipe | Depth: 2823 m | Formation: Sandstone | Severity: Medium | Duration: 11.4 hours

Well ID: W007 | Event Type: Casing Issue | Depth: 2852 m | Formation: Sandstone | Severity: High | Duration: 34.4 hours

7.1 EVENT DETAIL

EVENT E0040 — STUCK PIPE AT 2,317 m (SANDSTONE)

Well ID: W007
Event ID: E0040
Event Type: Stuck Pipe
Depth: 2317 m
Formation: Sandstone
Severity: High
Duration: 28.2 hours
Operational Impact: Significant non-productive time; remedial operations and additional materials required.

Description. Differential/mechanical sticking of the string at 2317 m in Sandstone; hole cleaning improved and pipe recovered. Parameters recorded around the event depth: mud weight 12.19 ppg, ROP 15.5 m/hr, 100 rpm, WOB 25.1 klbf.

Response. String worked and jarred within the allowable overpull; lubricant pill spotted and hole cleaning improved with high-viscosity sweeps and increased rotation before back-reaming.

Outcome. String freed and circulation restored. Hole conditioned over the affected interval and reaming carried out before drilling ahead.

Historical observation. Hole-cleaning efficiency and time spent stationary are the dominant contributors to sticking tendency in this section.

EVENT E0041 — STUCK PIPE AT 2,823 m (SANDSTONE)

Well ID: W007
Event ID: E0041
Event Type: Stuck Pipe
Depth: 2823 m
Formation: Sandstone
Severity: Medium
Duration: 11.4 hours
Operational Impact: Partial non-productive time; drilling suspended while the well was treated and conditioned.

Description. Differential/mechanical sticking of the string at 2823 m in Sandstone; hole cleaning improved and pipe recovered. Parameters recorded around the event depth: mud weight 12.94 ppg, ROP 13.7 m/hr, 108 rpm, WOB 28.3 klbf.

Response. String worked and jarred within the allowable overpull; lubricant pill spotted and hole cleaning improved with high-viscosity sweeps and increased rotation before back-reaming.

Outcome. String freed and circulation restored. Hole conditioned over the affected interval and reaming carried out before drilling ahead.

Historical observation. Hole-cleaning efficiency and time spent stationary are the dominant contributors to sticking tendency in this section.

EVENT E0042 — CASING ISSUE AT 2,852 m (SANDSTONE)

Well ID: W007
Event ID: E0042
Event Type: Casing Issue
Depth: 2852 m
Formation: Sandstone
Severity: High
Duration: 34.4 hours
Operational Impact: Significant non-productive time; remedial operations and additional materials required.

Description. Casing integrity concern noted opposite Sandstone at 2852 m; pressure test anomaly investigated before proceeding. Parameters recorded around the event depth: mud weight 12.97 ppg, ROP 13.6 m/hr, 105 rpm, WOB 28.6 klbf.

Response. Casing run suspended, string pulled back and the hole conditioned/reamed over the restriction before the casing was rerun; pressure test repeated after landing.

Outcome. Casing landed at the planned shoe depth and tested satisfactorily. Running speed limited through the affected interval on subsequent strings.

Historical observation. Tight/plastic intervals reduce casing running clearance; conditioning trips before running casing are effective.

8. CASING AND CEMENTING RECORD

Casing	Grade / Weight	Shoe Depth (m)	Hole Size	Cement Programme
13-3/8"	K-55, 68 lb/ft	765	17-1/2"	Lead 12.6 ppg / Tail 15.8 ppg, cement to surface
9-5/8"	N-80, 47 lb/ft	1,840	12-1/4"	Lead 12.9 ppg / Tail 15.8 ppg, TOC 650 m
7"	N-80, 29 lb/ft	3,400	8-1/2"	Tail 15.8 ppg, TOC 1,690 m

8.1 Cementing Observations

The following casing/cementing events were recorded and are cross-referenced to Section 7:

- **E0042** — Casing Issue at 2,852 m in Sandstone (High, 34.4 hr). Casing integrity concern noted opposite Sandstone at 2852 m; pressure test anomaly investigated before proceeding.

Remedial work was carried out as described in Section 7 and annular isolation was confirmed before drilling ahead.

8.2 Pressure Testing

Each string was pressure tested after landing and formation integrity tests were carried out below the 13-3/8" and 9-5/8" shoes. Recorded results were satisfactory except where noted in Section 7.

9. FINAL SUMMARY

Final Depth	3,400 m
Primary Formation	Sandstone
Major Events	2 of 3 events rated High or Critical
Total NPT	74.0 hours
Completion Status	Completed
Offset Risk Classification	HIGH — repeated significant events

9.1 Drilling Challenges

- **Stuck Pipe** at 2,317 m, 2,823 m (Sandstone) — 39.6 hr cumulative.
- **Casing Issue** at 2,852 m (Sandstone) — 34.4 hr cumulative.

9.2 Key Historical Observations

- Hole-cleaning efficiency and time spent stationary are the dominant contributors to sticking tendency in this section.
- Tight/plastic intervals reduce casing running clearance; conditioning trips before running casing are effective.
- Formation tops in this well correlate closely with the neighbouring wells listed in Section 4, supporting depth-based offset comparison across the cluster.

9.3 Lessons Learned

- Improve hole cleaning and minimise stationary time between 2,317 and 2,823 m; plan a wiper trip before extended surveys in this interval.
- Allow a dedicated conditioning trip before running casing across 2,852–2,852 m and limit running speed through the tight interval.

10. DOCUMENT CONTROL

Document Number	WCR-W007-R0
Revision	R0
Issue Date	04-Jul-2021
Prepared By	Drilling Engineering (prototype dataset generator)
Classification	Prototype Training Record — synthetic demonstration data
Source Datasets	wells.csv, well_formations.csv, drilling_events.csv, drilling_parameters.csv

All identifiers, coordinates, depths, formations, events and measurements in this document are synthetic and were generated for the eRTMAC-NWIS prototype. This document is not an operational record of Oil India Limited or any other operator, and no confidential information is contained in it.